Problem 9

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Problem 1 *Consider the parabola* $f(x) = ax^2 + bx + c$ *where* a, b, c *are constants. For what values of* a, b, c *is* f *concave up? For what values of* a, b, c *is* f *concave down?*

Explanation.

$$f'(x) = 2ax + b$$
$$f''(x) = 2a$$

This means the sign of a will determine whether f is concave up or down. When $a < 0$, f is concave down, which makes sense because then the graph of f is a downward opening parabola. When $a > 0$, f is concave up, which makes sense because then the graph of f is an upward openng parabola. If $a = 0$, f has no concavity because it is a linear function.